jupyter Kmeans++ clustering_mallpurchase_prediction Last Checkpo

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Titilt: Enhancing Retail Intelligence: A Customer Recommendations

The Business Challenge Traditional marketing often relies on a "broad-spectrum creates two major financial leaks:

Wasted Ad Spend: Marketing high-ticket luxury items to low-income or price-se

Profit Margin Erosion: Offering unnecessary discounts to "high-spending" loyalis

Currently, the business cannot distinguish between a customer who wants to bu This lack of insight results in missed sales opportunities and a poor return on ad

The Solution is ML model Recommendation system here I have choosen Kmea

K-Means++ Machine Learning model to automatically categorize customers bas

By grouping customers into mathematically precise clusters, the model identifies

Target Segments: Exactly which group should receive ads for premium vs. budge

Behavioral Trends: Who is likely to respond to a "value-driven" ad versus an "exc Recommendation. By ensuring the right product reaches the right target custom irrelevant ad placements

Note on Methodology: Multi-Phase Clustering Approa

To ensure both mathematical accuracy and visual interpretability, the model was